# Capacity-Radius Proximity for Smooth-Domain Reed–Solomon Codes

Chloe van der Vlugt

**Abstract**

We study list decoding and mutual correlated agreement for constant-rate Reed–Solomon codes over smooth evaluation domains. The primary received-word space is the two-fold interleaving $\mathcal{C}^{\equiv 2} \subseteq (\mathbb{F}^2)^n$. We isolate three layers that are often conflated: normalized distance of a received word from the interleaved code, worst-case list size at that radius, and the fraction of affine challenges for which a scalar projection appears close while its underlying pair does not. A field-valued two-coordinate plane is exactly an arbitrary word in $(\mathbb{F}^2)^n$, and it is equivalently the base–direction pair of an affine line of scalar received words. This gives a lossless interface between interleaved list decoding and mutual correlated agreement.

At count radius $n-k$, a direct agreement-set injection and a matching received-word construction prove

$$L_m\left(\mathcal{C}, \frac{n-k}{n}\right) = \binom{n}{k}$$

for every positive interleaving width. Thus $\binom{n}{k} \leq \varepsilon|\mathbb{F}|$ is necessary and sufficient for the requested fractional list target at the capacity endpoint. A next-shell construction then gives a complete three-regime phase split. We instantiate the safe regime with an explicit rate-1/4 smooth family over $\mathbb{F}_{3^n}$ satisfying $\binom{n}{k} \leq 2^{-128}|\mathbb{F}|$. In that regime the real safe-radius set is half-open and has a supremum but no largest real element. The optimal sub-capacity radius in the intermediate field-size regime and the capacity-radius mutual-correlated-agreement estimate remain separate proof targets.

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# 1 Introduction

Reed–Solomon proximity questions connect three quantitatively different objects:

(i) the relative distance of a received word from a code;

(ii) the number of codewords in a ball at that radius;

(iii) the fraction of affine challenges that conceal distance of an underlying tuple of received words.

The Grand List Decoding and Grand Mutual Correlated Agreement challenges ask for sharp forms of the latter two quantities for constant-rate Reed–Solomon codes over smooth evaluation domains [ABF26].

This paper treats the coding problem independently of any protocol or application-specific source of received words. Its universal statements quantify over every word in the ambient alphabet and every affine line. A later application may map structured observations into this ambient space, but that map is not part of the coding theorem.

### 1.1 Central organization

Let

$$\mathcal{C} = \mathrm{RS}[\mathbb{F}, \mathcal{L}, k] \subseteq \mathbb{F}^n, \qquad n = |\mathcal{L}|, \qquad \rho = \frac{k}{n}.$$

The two-fold interleaving is

$$\mathcal{C}^{\equiv 2} \subseteq (\mathbb{F}^2)^n.$$

Every pair word $p \in (\mathbb{F}^2)^n$ determines an affine line

$$y_\gamma(p) = p_0 + \gamma p_1 \in \mathbb{F}^n, \qquad \gamma \in \mathbb{F}.$$

The intended proof architecture is therefore

$$\begin{array}{ccc} p \in (\mathbb{F}^2)^n & \xrightarrow{\gamma \in \mathbb{F}} & y_\gamma(p) \in \mathbb{F}^n \\ \text{distance and list size relative to } \mathcal{C}^{\equiv 2} & & \text{distance relative to } \mathcal{C}. \end{array}$$

List decoding controls ambiguity of the full pair; MCA controls the fraction of challenges for which a scalar observation fails to expose pair-farness.

### 1.2 Claim discipline

The plane/pair/line identifications below are exact equivalences. They ensure that no received words or affine lines are lost by the chosen presentation. They do not by themselves prove capacity-radius list decoding or MCA. The list theorem is supplied separately by an agreement-set injection, a matching endpoint witness, and an explicit field-size inequality. MCA remains a distinct quantitative target.

In particular, a statement of the form

$$\operatorname{dist}(p, \mathcal{C}^{\equiv 2}) \geq \delta$$

is not a list-size bound, and neither fact alone implies a security estimate of $2^{-128}$. The required list size or bad-challenge count must be compared explicitly with $\varepsilon^*|\mathbb{F}|$.

### 1.3 Contributions sought

This draft establishes:

(i) the universal capacity-grid equality $L_m(\mathcal{C}, (n-k)/n) = \binom{n}{k}$;

(ii) the necessary-and-sufficient endpoint criterion $\binom{n}{k} \leq \varepsilon^*|\mathbb{F}|$ and the resulting three-regime phase split;

(iii) a next-shell obstruction of size $|\mathbb{F}|^m$;

(iv) an explicit smooth rate-1/4 family and field-size inequality for $\varepsilon^* = 2^{-128}$;

(v) machine-checked definitions, reductions, both endpoint inequalities, their exact fractional-target equivalence, and the family arithmetic.

The remaining goals are the mutual-correlated-agreement estimate, the application calibration theorem, and the optimal sub-endpoint radius in the intermediate field-size regime.

## 2 Problem Setup

### 2.1 Reed–Solomon parameters

Let $\mathbb{F}$ be a finite field and let $\mathcal{L} = (a_i)_{i \in I}$ be an injectively indexed evaluation domain, where $|I| = n$. For $1 \leq k \leq n$, define

$$\mathcal{C} = \mathrm{RS}[\mathbb{F}, \mathcal{L}, k] = \{(f(a_i))_{i \in I} : f \in \mathbb{F}[X],\ \deg f < k\}\,.$$

The rate is $\rho = k/n$. The prize-facing regime takes $\rho$ from a fixed constant set, such as

$$\left\{\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}\right\},$$

and requires $\mathcal{L}$ to have an appropriate smooth structure.

### 2.2 Relative Hamming distance

For a finite alphabet $A$ and $u, v \in A^I$, let

$$d_H(u,v) = |\{i \in I : u_i \neq v_i\}|, \qquad \Delta(u,v) = \frac{d_H(u,v)}{n}.$$

For a code $\mathcal{D} \subseteq A^I$, define

$$\Delta(w, \mathcal{D}) = \min_{c \in \mathcal{D}} \Delta(w,c).$$

All radii in the main theorem statements are relative radii in $[0,1]$. Natural-number radii remain useful for finite rounding, but they are not the paper-facing metric.

### 2.3 Interleaving

For a positive integer $m$, define

$$\mathcal{C}^{\equiv m} = \left\{ p \in (\mathbb{F}^m)^I : \text{for every } j \in \{1,\dots,m\},\ (p_{i,j})_{i \in I} \in \mathcal{C} \right\}.$$

Hamming distance on $(\mathbb{F}^m)^I$ counts a coordinate $i$ as erroneous when the complete vector symbol differs. It does not sum componentwise scalar distances.

### 2.4 List-decoding balls

For $w \in (\mathbb{F}^m)^I$ and $\delta \in [0,1]$, define

$$\Lambda(\mathcal{C}^{\equiv m}, w, \delta) = \{ c \in \mathcal{C}^{\equiv m} : \Delta(w,c) \leq \delta \}.$$

The worst-case list size is

$$L_m(\mathcal{C}, \delta) = \max_{w \in (\mathbb{F}^m)^I} |\Lambda(\mathcal{C}^{\equiv m}, w, \delta)|.$$

The maximum over received words is essential. A bound on a structured subset of received words is an application result, not the universal list-decoding statement.

# 3 Planes, Pair Words, And Affine Lines

## 3.1 The full ambient pair space

Define a field-valued two-coordinate plane to be a function

$$p : I \longrightarrow (\{0,1\} \longrightarrow \mathbb{F}).$$

By reassociating function spaces, the plane carrier is exactly

$$I \to (\{0,1\} \to \mathbb{F}) \cong (\mathbb{F}^2)^I.$$

Thus the word "plane" introduces no restriction on the received-word space.

**Proposition 3.1** (Plane–pair equivalence)**.** *The map sending a plane $p$ to the vector word*

$$i \longmapsto (p(i,0), p(i,1))$$

*is a bijection from field-valued planes to $(\mathbb{F}^2)^I$. It preserves and reflects Hamming distance.*

*Proof.* The inverse sends a pair word $w$ to the function

$$(i,j) \longmapsto w_i(j).$$

Both composites are pointwise identities. At coordinate $i$, the two planes are equal exactly when their complete pair symbols are equal, so the Hamming disagreement sets coincide. □

## 3.2 Pair words as affine lines

For a plane $p$, define its base and direction by

$$p_0(i) = p(i,0), \qquad p_1(i) = p(i,1).$$

The associated affine line of scalar received words is

$$\ell_p(\gamma) = p_0 + \gamma p_1, \qquad \gamma \in \mathbb{F}.$$

**Proposition 3.2** (Plane–line equivalence)**.** *The map $p \mapsto (p_0, p_1)$ is a bijection between field-valued planes and affine-line presentations consisting of an arbitrary base word and arbitrary direction word in $\mathbb{F}^I$.*

*Proof.* Given an arbitrary pair $(f_0, f_1)$, define

$$p(i,0) = f_0(i), \qquad p(i,1) = f_1(i).$$

This construction is inverse to taking base and direction. □

**Remark 3.3.** *These equivalences discharge the coverage question: a theorem quantified over all planes is a theorem over every received word in $(\mathbb{F}^2)^I$ and every affine-line presentation. No surjectivity claim from an application-specific frame or commitment space is required.*

### 3.3 Challenge evaluation is lossy

For fixed $\gamma$, the map

$$(p_0, p_1) \longmapsto p_0 + \gamma p_1$$

is generally many-to-one. Consequently, pair-farness cannot be inferred deterministically from one fixed challenge without an additional good-challenge hypothesis. MCA supplies a probabilistic statement over a challenge sampled after the pair has been fixed.

## 4 The Capacity-Radius Regime

### 4.1 The finite Hamming grid

Let $1 \le k \le n$. The capacity endpoint is already an attainable normalized Hamming radius:

$$r_{\mathrm{cap}} = n - k, \qquad \delta_{\mathrm{cap}} = \frac{n-k}{n} = 1 - \rho.$$

There is therefore no rounding loss at the endpoint. For example, the rate-$1/4$, length-4096 member has

$$k = 1024, \qquad r_{\mathrm{cap}} = 3072, \qquad \delta_{\mathrm{cap}} = \frac{3}{4}.$$

The previously useful strict checkpoint $3071/4096$ is subsumed by the endpoint theorem below.

There is, however, a distinct granularity issue in the phrase "largest relative radius." A closed Hamming ball changes only at

$$\mathcal{H}_n = \left\{0, \frac{1}{n}, \ldots, \frac{n}{n}\right\}.$$

Consequently, a safe-radius set defined over all real $\delta \in [0, 1]$ may be half-open and have no largest element. We report both the largest radius in $\mathcal{H}_n$ and the supremum over real radii.

### 4.2 An explicit smooth large-field family

For $u \geq 0$, set

$$k_u = 64 \cdot 2^u = 2^{u+6}, \qquad n_u = 4k_u = 256 \cdot 2^u = 2^{u+8}, \qquad q_u = 3^{n_u}.$$

Let $\mathbb{F}_u = \mathbb{F}_{q_u}$. Since the multiplicative group of a finite field is cyclic of order $q_u - 1$, it suffices to check $n_u \mid (q_u - 1)$. Euler's theorem gives this directly:

$$\varphi(n_u) = \frac{n_u}{2}, \qquad \gcd(3, n_u) = 1, \qquad 3^{n_u/2} \equiv 1 \pmod{n_u}.$$

Squaring yields $3^{n_u} \equiv 1 \pmod{n_u}$. Thus $\mathbb{F}_u^\times$ contains a subgroup $H_u$ of order $n_u$. We take $\mathcal{L}_u = H_u$ and

$$\mathcal{C}_u = \mathrm{RS}[\mathbb{F}_u, H_u, k_u].$$

The evaluation length and degree bound are powers of two, the domain is a multiplicative subgroup, and the rate is exactly $1/4$. This is the standard smooth-domain condition for FFT-based Reed–Solomon codes [ACFY24], and it is the explicit family used in the quantitative statement.

### 4.3 The field-size inequality

Fix $\varepsilon^* = 2^{-128}$. For the family above,

$$\binom{n_u}{k_u} \leq 2^{n_u}.$$

Moreover $n_u$ is even and $n_u/2 \geq 128$, so

$$\begin{aligned} q_u = 3^{n_u} &= (3^2)^{n_u/2} = 9^{n_u/2} \\ &\geq 8^{n_u/2} = 2^{3n_u/2} \geq 2^{n_u+128}. \end{aligned}$$

It follows that

$$\binom{n_u}{k_u} \leq 2^{-128} q_u = \varepsilon^* |\mathbb{F}_u|.$$

The field is exponential in the block length. This is a valid explicit sufficiently-large-field regime, not a claim about practically sized fields or about every smooth code supplied to the Grand Challenge.

## 5 The Capacity-Grid List Theorem

The following argument applies to every injectively indexed Reed–Solomon evaluation domain; smoothness enters only when we instantiate the explicit family from Section 4.2.

**Theorem 5.1** (Universal capacity-grid equality)**.** *Let $\mathcal{C} = \mathrm{RS}[\mathbb{F}, \mathcal{L}, k]$ have length $n$, where $1 \le k \le n$, and let $m \ge 1$. Then*

$$L_m\left(\mathcal{C}, \frac{n-k}{n}\right) = \binom{n}{k}.$$

*In particular, the worst-case endpoint list size is independent of $m$.*

*Proof.* We first prove the upper bound. Fix $w \in (\mathbb{F}^m)^n$. Every codeword in the ball agrees with $w$ on at least $k$ complete vector-symbol coordinates. Assign to each such codeword one $k$-subset of its agreement coordinates. This assignment is injective: if two interleaved codewords are assigned the same set $S$, then they agree with one another on $S$. In each component, two degree-$< k$ polynomials agree at the $k$ distinct evaluation points indexed by $S$, and hence are equal. The two interleaved codewords are therefore equal. There are exactly $\binom{n}{k}$ possible sets $S$.

For the reverse inequality, choose one of the $m$ components and define the received word

$$w_i^\star = (a_i^k, 0, \dots, 0).$$

For each $k$-subset $S \subseteq I$, put

$$V_S(X) = \prod_{i \in S}(X - a_i), \qquad f_S(X) = X^k - V_S(X).$$

The polynomial $V_S$ is monic of degree $k$, so its leading term cancels that of $X^k$ and $\deg f_S < k$. Use $f_S$ in the distinguished component and zero polynomials in all other components to obtain an interleaved codeword $c_S$.

At evaluation coordinate $i$,

$$c_{S,i} = w_i^\star \quad \Longleftrightarrow \quad V_S(a_i) = 0 \quad \Longleftrightarrow \quad i \in S,$$

where the final equivalence uses injectivity of the evaluation points. Consequently $c_S$ has distance exactly $n-k$ from $w^\star$, and its agreement set recovers $S$. The $\binom{n}{k}$ choices of $S$ therefore give distinct codewords in the endpoint ball. This matches the upper bound. □

**Theorem 5.2** (Failure on the next Hamming shell)**.** *Under the same hypotheses,*

$$\left|\Lambda\left(\mathcal{C}^{\equiv m}, 0, \frac{n-k+1}{n}\right)\right| \geq |\mathbb{F}|^m.$$

*Proof.* Fix $k-1$ evaluation points $S \subseteq \mathcal{L}$ and set

$$g_S(X) = \prod_{a \in S}(X - a).$$

For every $\lambda = (\lambda_1, \ldots, \lambda_m) \in \mathbb{F}^m$, the tuple

$$(\lambda_1 g_S, \ldots, \lambda_m g_S)$$

defines an interleaved codeword. These $|\mathbb{F}|^m$ codewords are distinct. The zero tuple has distance zero from the received word 0; every nonzero tuple vanishes on $S$ and is nonzero at every evaluation point outside $S$, so its distance is exactly $n-k+1$. □

**Theorem 5.3** (Fractional-target phase split)**.** *Fix $0 < \varepsilon^* < 1$, write $q = |\mathbb{F}|$, and call a radius $\delta$ safe when*

$$L_m(\mathcal{C}, \delta) \leq \varepsilon^* q.$$

*For every $m \geq 1$, exactly one of the following regimes applies:*

*(i) If $\varepsilon^* q < 1$, no radius in $[0,1]$ is safe.*

*(ii) If $1 \leq \varepsilon^* q < \binom{n}{k}$, radius zero is safe but the capacity endpoint and every larger radius fail. The optimal sub-endpoint radius is not determined by the present argument.*

*(iii) If $\binom{n}{k} \leq \varepsilon^* q$, the largest safe radius on $\mathcal{H}_n$ is*

$$\delta_{\text{cap}} = \frac{n-k}{n} = 1 - \rho;$$

*and over real radii the safe set is exactly*

$$\left[0, \frac{n-k+1}{n}\right),$$

*so it has no largest element and has supremum $(n-k+1)/n$.*

*Equivalently, the fractional target holds at the capacity endpoint if and only if*

$$\binom{n}{k} \leq \varepsilon^* |\mathbb{F}|.$$

*Proof.* Every codeword is contained in the radius-zero ball centered at itself, so no radius can be safe when $\varepsilon^* q < 1$. A radius-zero ball contains at most one codeword, proving its safety when $\varepsilon^* q \geq 1$.

Theorem 5.1 proves both the failure in the intermediate regime and the necessary-and-sufficient endpoint criterion. In the final regime, it and monotonicity of list balls prove safety through count radius $n - k$. A real radius strictly below $(n-k+1)/n$ defines the same ball as some count radius at most $n - k$. At the next grid point, Theorem 5.2 gives a list of size at least $|\mathbb{F}|^m \geq |\mathbb{F}| > \varepsilon^*|\mathbb{F}|$; monotonicity then rules out every larger radius. □

**Corollary 5.4** (The explicit smooth family)**.** *For every $u \geq 0$, every $m \geq 1$, and $\mathcal{C}_u = \mathrm{RS}[\mathbb{F}_u, H_u, k_u]$ from Section 4.2, the conclusions of Theorem 5.3*(iii) *hold with $\varepsilon^* = 2^{-128}$. In particular, the largest safe Hamming-grid radius is* $3/4$.

*Proof.* Apply the inequality proved in Section 4.3. □

### 5.1 Scope relative to the Grand Challenge

Theorem 5.1 completely determines the endpoint list size for every supplied Reed–Solomon code, smooth or otherwise. Thus the displayed binomial field-size inequality is necessary and sufficient at that endpoint, not merely a convenient sufficient hypothesis. Theorem 5.4 supplies a concrete smooth family in the safe regime. The present argument does not determine the largest sub-endpoint radius for an arbitrary fixed smooth code in the intermediate regime

$$1 \leq 2^{-128}|\mathbb{F}| < \binom{n}{k}.$$

The exponential alphabet is what places the explicit family in the final regime; it is not needed for the universal endpoint equality.

This distinction matters in light of known capacity results. Generic Reed–Solomon evaluation points can achieve list-decoding capacity over exponentially large fields with constant list size [BGM24], but generic points do not supply the structured smooth domain required here. Conversely, near-capacity proximity-gap behavior can fail for particular smooth prime-field families [Kam26]. The theorem above imports neither result: it uses only polynomial interpolation and an explicit field-size comparison.

# 6 The Mutual-Correlated-Agreement Target

## 6.1 Bad challenges

Let $p = (p_0, p_1) \in (\mathbb{F}^2)^I$, and write

$$y_\gamma = p_0 + \gamma p_1.$$

For $S \subseteq I$, say that $S$ is $\delta$-large when

$$\frac{|S|}{n} \geq 1 - \delta.$$

Define $\mathrm{Bad}_{\mathrm{MCA}}(\mathcal{C}, \delta, p)$ to be the set of $\gamma \in \mathbb{F}$ for which there exists a $\delta$-large set $S$ such that:

(i) $y_\gamma$ agrees on $S$ with a codeword of $\mathcal{C}$;

(ii) the pair $p$ does not agree on the same $S$ with a codeword of $\mathcal{C}^{\equiv 2}$.

The MCA error is

$$\varepsilon_{\mathrm{MCA}}(\mathcal{C}, \delta) = \max_{p \in (\mathbb{F}^2)^I} \frac{|\,\mathrm{Bad}_{\mathrm{MCA}}(\mathcal{C}, \delta, p)|}{|\mathbb{F}|}.$$

**Problem 6.1** (Capacity-radius MCA bound)**.** *For a constant-rate smooth-domain Reed–Solomon code, determine an explicit bound*

$$B_{\mathrm{MCA}}(\mathbb{F}, n, k)$$

*such that*

$$\forall p \in (\mathbb{F}^2)^n, \qquad |\,\mathrm{Bad}_{\mathrm{MCA}}(\mathcal{C}, \delta_{\mathrm{cap}}, p)| \leq B_{\mathrm{MCA}}(\mathbb{F}, n, k),$$

*and determine when*

$$\frac{B_{\mathrm{MCA}}(\mathbb{F}, n, k)}{|\mathbb{F}|} \leq \varepsilon^*.$$

## 6.2 Distance interpretation

Let

$$D_2(p) = \Delta(p, \mathcal{C}^{\equiv 2}), \qquad D_1(y) = \Delta(y, \mathcal{C}).$$

For a challenge outside the MCA bad set, scalar closeness on a large agreement set forces pair closeness on that same set. In distance language, the desired contrapositive has the form

$$D_2(p) \geq \delta \Longrightarrow D_1(y_\gamma) \geq \delta$$

for every MCA-good $\gamma$, with the fraction of excluded challenges bounded by $\varepsilon_{\mathrm{MCA}}(\mathcal{C}, \delta)$.

This is a challenge-indexed statement. Replacing the sampled challenge by one fixed field element would require a separate theorem that the selected element is good for every pair, which does not follow from a fractional MCA bound.

### 6.3 Commit-then-challenge order

The mathematical experiment fixes $p$ before sampling $\gamma \leftarrow \mathbb{F}$. This ordering prevents the pair from being selected as a function of the challenge. A full protocol implementation must bind $p$ before deriving $\gamma$, but the coding theorem needs only the resulting quantification and probability space; it does not depend on a complete blockchain or commitment protocol.

### 6.4 Proof status

[TODO: Insert the capacity-radius MCA argument and its explicit field-size bound.]

## 7 Application Interface

The coding theorem is stated over the complete received-word space and does not depend on a particular source of observations. An application may nevertheless inherit the result through a distance-aware map.

Let $O$ be a type of raw presented observations, let $G \subseteq O$ be the exact/valid locus, and let

$$\pi : O \longrightarrow (\mathbb{F}^2)^I$$

be the extracted pair word. A source distance $\rho_O : O \to [0, 1]$ is meaningful only when $O$ includes arbitrary or noisy presentations; if every element of $O$ already contains a validity witness, then the proximity problem may collapse to the exact code locus.

The principal application theorem should have one of the forms

$$D_2(\pi(o)) = \rho_O(o),$$

or

$$\rho_O(o) \geq \delta \Longrightarrow D_2(\pi(o)) \geq \delta.$$

Under either comparison, the proved universal list theorem applies to $\pi(o)$; a future universal MCA theorem would apply through the same interface.

### 7.1 Exact witnesses versus raw observations

The recommended type boundary is:

| | |
|---|---|
| raw observation | arbitrary public presentation, |
| exact witness | certificate of valid structured realization, |
| $G(o)$ | there exists an exact witness realizing $o$. |

An exact witness is evidence for membership in the code locus; it should not be the carrier of adversarial received words.

### 7.2 Separation from application papers

Application-specific Plucker coordinates, cochain commitments, hidden operators, transcript binding, and runtime verification are intentionally outside the proof of Theorems 5.1, 5.2 and 6.1. They may cite the completed list theorem, and eventually the MCA theorem, through the interface above. Conversely, each coding theorem must remain valid without assuming that every received word comes from an application-specific witness space.

## 8 Formalization And Proof Ledger

### 8.1 Standalone hierarchy

The intended machine-checked development should separate generic coding mathematics from application bridges:

`ProximityPrize/BlockCode`
: Relative distance, balls, and worst-case list size.

`ProximityPrize/ReedSolomon`
: Smooth-domain RS families and interleaving.

`ProximityPrize/PlanePair`
: Plane–pair–line equivalences.

`ProximityPrize/Parameters`
: Explicit $\varepsilon^*$ and field-size inequalities.

`ProximityPrize/CapacityList`
: Agreement-set injection, the matching received-word construction, the universal endpoint equality, and the fractional-target equivalence.

Existing generic definitions may be imported or refactored, but the final theorem should not depend on HCP, protocol, consensus, or runtime axiom bundles. Both directions of the capacity-grid equality and its exact fractional-target criterion are theorems, not axioms. The next-shell lower bound and the capacity-MCA target are not declared as Lean theorems or axioms at this stage.

### 8.2 Status ledger

| Obligation | Status | Required endpoint |
|---|---|---|
| Relative Hamming normalization | Lean-defined | Values in $[0, 1]$ |
| Plane–pair equivalence | Lean-proved | Bijection and Hamming isometry |
| Plane–affine-line equivalence | Lean-proved | Formal inverse maps |
| Smooth large-field family | Paper-proved | Explicit subgroup family |
| Capacity-grid list equality | Lean-proved | Universal $L_m = \binom{n}{k}$ |
| Endpoint fractional criterion | Lean-proved | Target iff $\binom{n}{k} \leq \varepsilon^* \lvert\mathbb{F}\rvert$ |
| Next-shell list lower bound | Paper-proved | Explicit $\lvert\mathbb{F}\rvert^m$ witness |
| Capacity-radius MCA bound | Open | Universal bad-set theorem |
| Target-fraction inequality | Lean-proved | Bound $\leq \varepsilon^* \lvert\mathbb{F}\rvert$ |
| Largest grid radius / real supremum | Paper-proved | Exact characterization |

### 8.3 Submission boundary

A completed claim requires:

(i) no placeholder assumptions for the main list/MCA result;

(ii) an explicit statement of imported literature theorems;

(iii) all field-size and rounding conditions checked;

(iv) a clear separation between proved theorems, computational evidence, and conjectures;

(v) a reproducible proof artifact and manuscript cross-reference.

### 8.4 Immediate next step

The endpoint list-decoding package is closed. Two logically separate tasks remain. The application side needs a quantitative non-collapse/calibration theorem for an extraction map $\pi$:

$$d_{\text{source}}(x, \text{valid}) \geq \delta \Longrightarrow \Delta(\pi(x), \mathcal{C}^{\equiv 2}) \geq \delta.$$

The remaining prize-facing coding task is the capacity-radius MCA estimate.

## References


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